

ASSIGNMENT: 4

1. Check if a Number is a Palindrome

Algorithm:

1. Start
2. Input num
3. Set original = num, reverse = 0
4. Repeat while num > 0
 - Get last digit: digit = num % 10
 - reverse = reverse * 10 + digit
 - num = num / 10
5. If original == reverse → Print "Palindrome"
6. Else → Print "Not a palindrome"
7. End

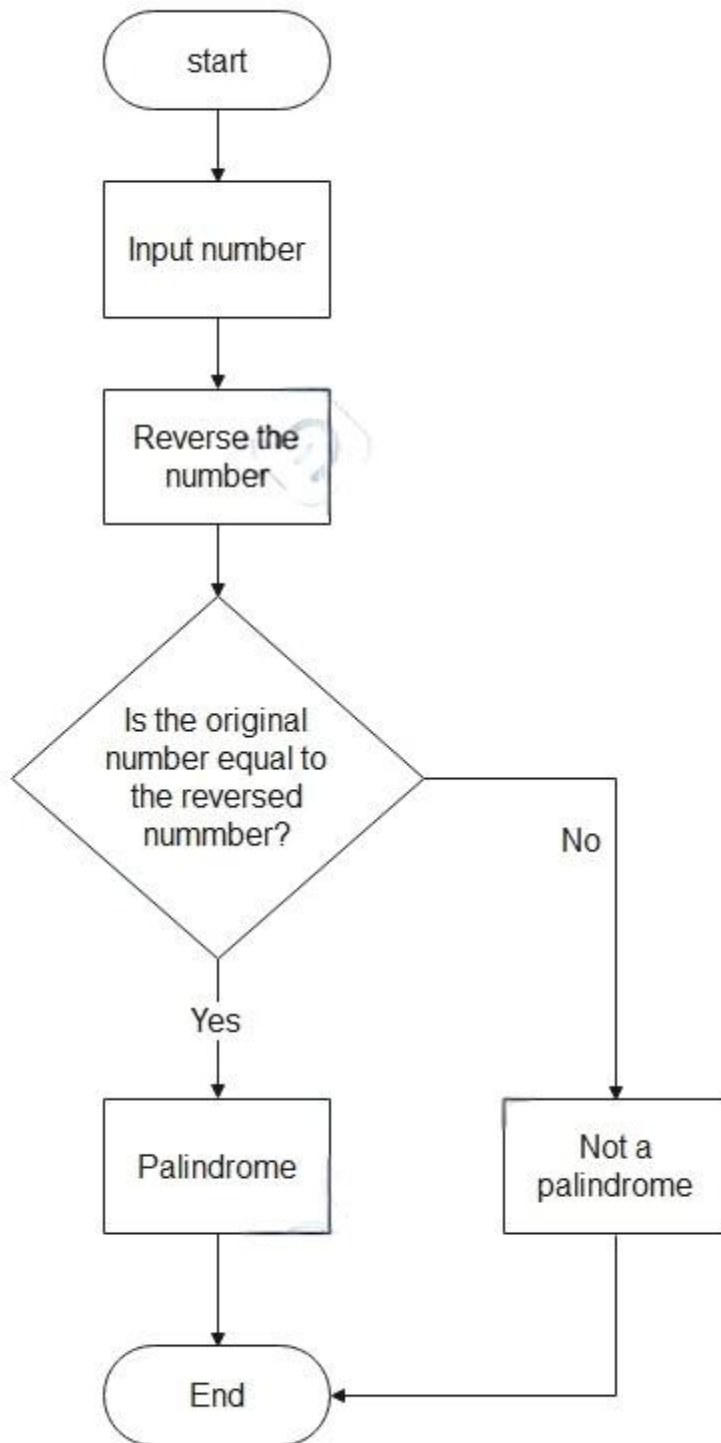
Code:

```
#include <stdio.h>
```

```
int main() {  
    int num, original, reverse = 0, digit;  
    printf("Enter a number: ");  
    scanf("%d", &num);  
    original = num;  
  
    while(num > 0) {  
        digit = num % 10;  
        reverse = reverse * 10 + digit;  
        num /= 10;  
    }  
  
    if(original == reverse)  
        printf("Palindrome\n");  
    else  
        printf("Not a palindrome\n");  
}
```

```
    return 0;  
}
```

Flowchart:



2. Sum of All Prime Numbers from 1 to n

Algorithm:

1. Start
2. Input n, initialize sum = 0
3. For i = 2 to n:
 - Set isPrime = 1
 - For j = 2 to i / 2
 - If i % j == 0 → Set isPrime = 0 and break
 - If isPrime == 1 → sum+=1
4. Print sum
5. End

Code:

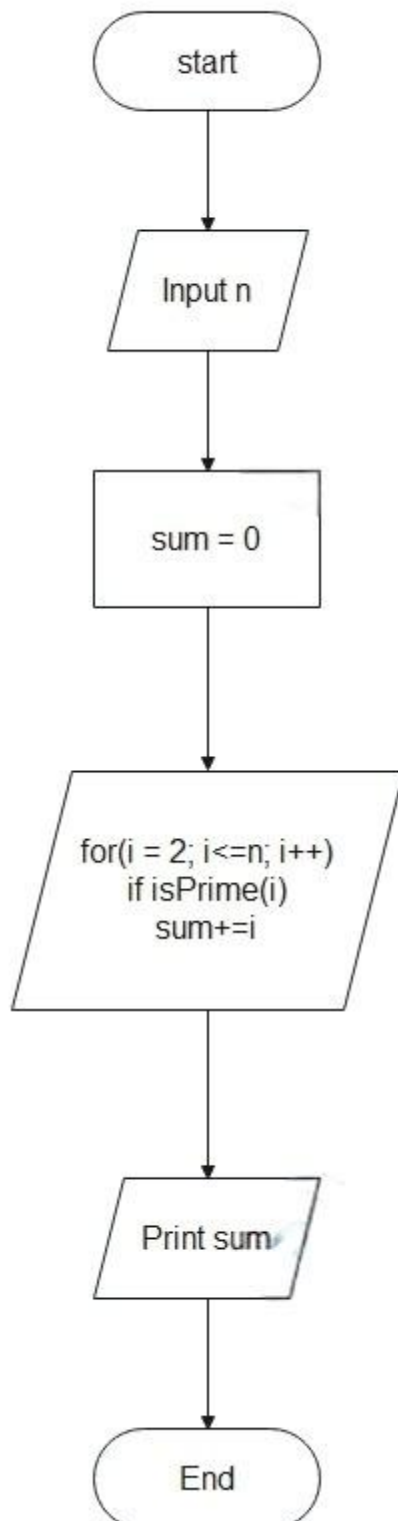
```
#include <stdio.h>
```

```
int main() {
    int n, i, j, isPrime, sum = 0;
    printf("Enter the value of n: ");
    scanf("%d", &n);

    for(i = 2; i <= n; i++) {
        isPrime = 1;
        for(j = 2; j <= i / 2; j++) {
            if(i % j == 0) {
                isPrime = 0;
                break;
            }
        }
        if(isPrime)
            sum += i;
    }

    printf("Sum of prime numbers = %d\n", sum);
    return 0;
}
```

Flowchart:



3. Check for a Perfect Number

Algorithm:

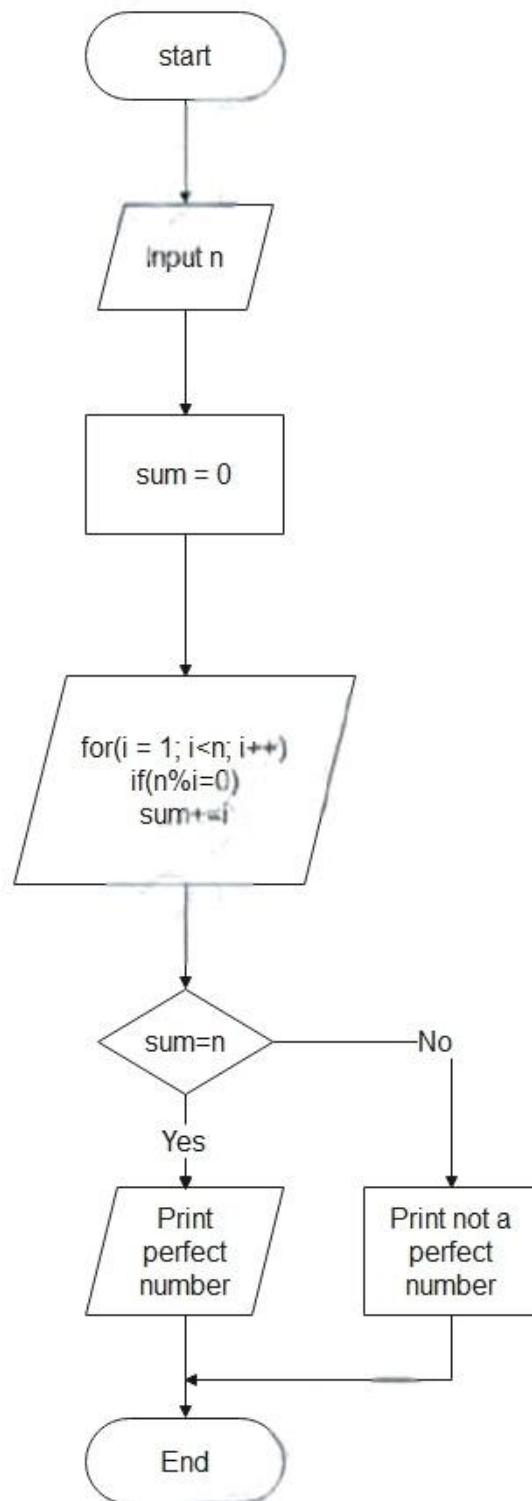
1. Start
2. Input num, initialize sum = 0
3. For i = 1 to num / 2
 - If num % i == 0 → sum += i
4. If sum == num → Print "Perfect number"
5. Else → Print "Not a perfect number"
6. End

Code:

```
#include <stdio.h>
```

```
int main() {  
    int num, i, sum = 0;  
    printf("Enter a number: ");  
    scanf("%d", &num);  
  
    for(i = 1; i <= num / 2; i++) {  
        if(num % i == 0)  
            sum += i;  
    }  
  
    if(sum == num)  
        printf("Perfect number\n");  
    else  
        printf("Not a perfect number\n");  
  
    return 0;  
}
```

Flowchart:



4. Fibonacci Series up to n Terms

Algorithm:

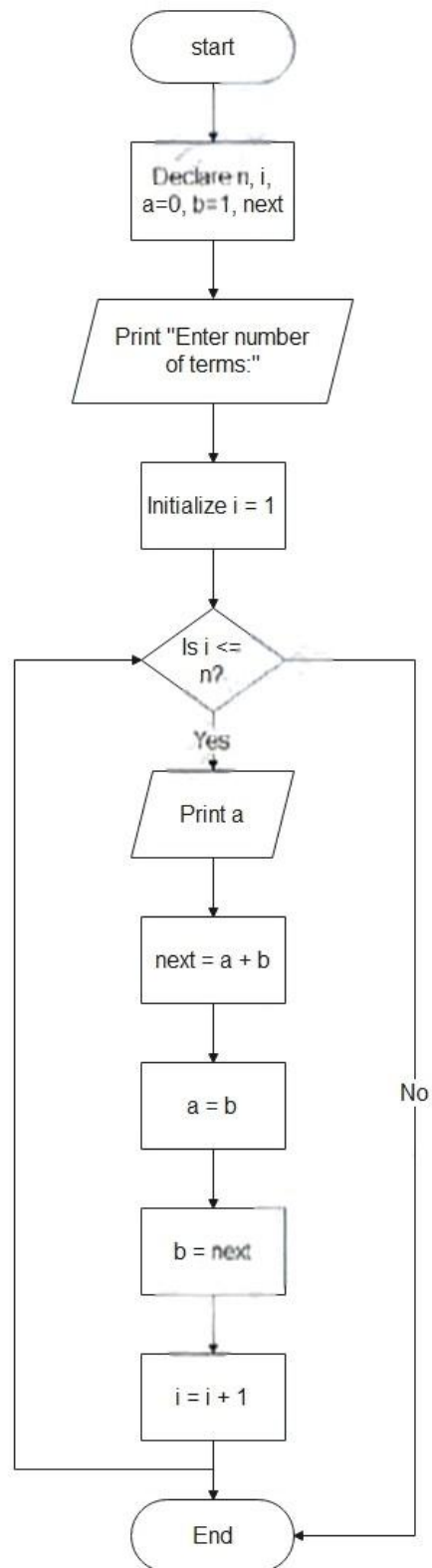
1. Start
2. Input n
3. Initialize: $a = 0$, $b = 1$
4. For $i = 1$ to n
 - Print a
 - $next = a + b$
 - $a = b$, $b = next$
5. End

Code:

```
#include <stdio.h>
```

```
int main() {  
    int n, i, a = 0, b = 1, next;  
    printf("Enter number of terms: ");  
    scanf("%d", &n);  
  
    for(i = 1; i <= n; i++) {  
        printf("%d ", a);  
        next = a + b;  
        a = b;  
        b = next;  
    }  
    return 0;  
}
```

Flowchart:



5. Armstrong Numbers between 1 to n

Algorithm:

1. Start
2. Input n
3. For i = 1 to n
 - Set num = i, sum = 0, count = 0, temp = num
 - Count digits:
 - While temp > 0 → count++, temp /= 10
 - Set temp = num
 - While temp > 0
 - digit = temp % 10
 - sum += pow(digit, count)
 - temp /= 10
 - If sum == num → Print num
4. End

Code:

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    int n, i, num, temp, sum, digit, count;
```

```
    printf("Enter the value of n: ");
```

```
    scanf("%d", &n);
```

```
    for(i = 1; i <= n; i++) {
```

```
        num = i;
```

```
        temp = num;
```

```
        count = 0;
```

```
        sum = 0;
```

```
        while(temp > 0) {
```

```
            temp /= 10;
```

```
            count++;
```

```
        }
```

```
        temp = num;
```

```
        while(temp > 0) {
```

```

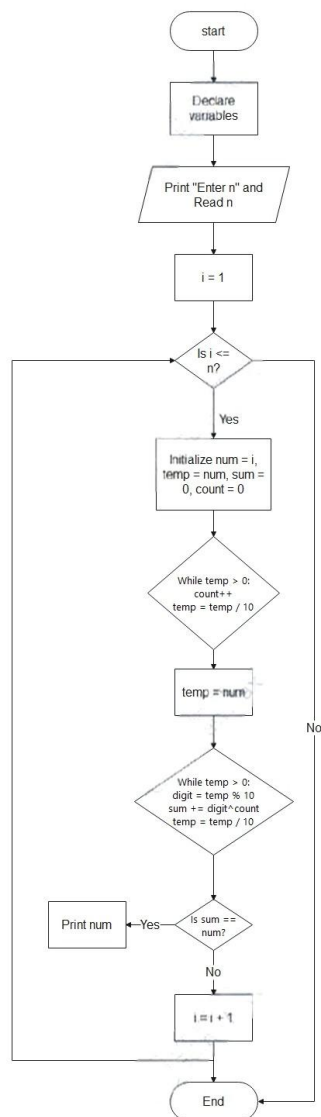
        digit = temp % 10;
        sum += pow(digit, count);
        temp /= 10;
    }

    if(sum == num)
        printf("%d ", num);
    }

return 0;
}

```

Flowchart:



6. Triangle Pattern of Prime Numbers

Algorithm:

1. Start
2. Input n
3. For i = 1 to n:
 - Get start = ith prime
 - For j = 1 to i:
 - Print start
 - start = next prime of start
4. End

Code:

```
#include <stdio.h>
```

```
int isPrime(int num) {  
    if(num < 2) return 0;  
    for(int i = 2; i <= num/2; i++)  
        if(num % i == 0)  
            return 0;  
    return 1;  
}
```

```
int nextPrime(int current) {  
    int next = current + 1;  
    while(!isPrime(next))  
        next++;  
    return next;  
}
```

```
int nthPrime(int n) {  
    int count = 0, num = 1;  
    while(count < n) {  
        num++;  
        if(isPrime(num))  
            count++;  
    }  
    return num;  
}
```

```
int main() {
```

```
int n, i, j;
printf("Enter the number of rows: ");
scanf("%d", &n);

for(i = 1; i <= n; i++) {
    int prime = nthPrime(i);
    for(j = 1; j <= i; j++) {
        printf("%d ", prime);
        prime = nextPrime(prime);
    }
    printf("\n");
}

return 0;
}
```

Flowchart:

